BT603 QUIZ# 2 → 6 JUNE,2023
TOPICS 25-34
BY TABIH FATIMA
ADMIN ALL ABOUT BIOTECH- AAB

To join my group email me at → vuallaboutbiotechofficial@gmail.com

Question # 9 of 10 (Start time: 08:37:48 AM, 07 June 2023)

_____ et al., (1984) discussed factors favouring the use of oils instead of carbohydrates

Select the correct option

- jack
- Bader
- bush
- jill

Click to Save Answer

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Question # 4 of 10 (Start time: 08:31:25 AM, 07 June 2023) **Total Marks:**

The concentration of _____ in a medium, particularly laboratory media in shake flasks, is often much higher than that of other mineral components.

Select the correct option

- Rose Bengal Agar
- phosphate
- Potato Dextrose Agar
- MacConkey's Agar

Click to Save Answer & Move to Next Question

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BT603 - Fermentation Technology (Quiz No. 2)

Quiz Start Time: 08:28 AM, 07 June 2023

Question # 6 of 10 (Start time: 08:34:30 AM, 07 June 2023)

Total Marks:

Most industrially used micro-organisms can utilize inorganic or organic sources of _____

Select the correct option

- hydrogen
- oxygen
- nitrogen
- helium



Click to Save Answer & Move to Next Question

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Question # 7 of 10 (Start time: 08:35:28 AM, 07 June 2023) Total

On a weight basis n-alkanes have approximately _____ the carbon and three times the energy conter the same weight of sugar.

Select the correct option

- five times
- thrice
- twice
- foure

Click to Save Answer & Move to Next Ques

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Question # 3 of 10 (Start time: 08:30:33 AM, 07 June 2023)

Some proteinaceous nitrogen compounds serving as sources of amino acids include corn-steep liquor, soya meal, peanut meal, _____.

Select the correct option

- carbon
- oxygen
- hydrogen
- cotton-seed meal



Click to Save Answer & Move to Nex

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BT603 - Fermentation Technology (Quiz No. 2) Quiz Start Time: 08:28 AM, 07 June 2023

Question # 2 of 10 (Start time: 08:29:52 AM, 07 June 2023) Total Marks:

There are some advantages and disadvantages of hydrocarbons and their derivatives as fermentation substrates, particularly with reference to _____, process aspects and purity.

Select the correct option

- cost
- system
- media
- air

Click to Save Answer & Move to Next Question

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Question # 1 of 10 (Start time: 08:28:50 AM, 07 June 2023)

Total Marks:

_____ is the major component of all fermentation media, and is needed in many of the ancillary services.

Select the correct option

- carbon
- hydrogen
- water
- nitrogen



Click to Save Answer & Move to Next Question

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Question # 8 of 10 (Start time: 08:18:57 AM, 07 June 2023)
Total Marks: 1

In griseofulvin production, adequate available chloride is provided by the inclusion of at least 0.1% _____ as well as the chloride provided by the complex organic materials included as nitrogen sources.

Select the correct option

- ○ KCl
- ○ Complex media
- ○ N
- ○ C

Click to Save Answer & Move to Next Question

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Time Left sec(s)

BT603 - Fermentation Technology (Quiz No. 2) Quiz Start Time: 08:12 AM, 07 June 202

Question # 6 of 10 (Start time: 08:17:25 AM, 07 June 2023) Total Marks:

_____ is being used as a buffer to minimize pH changes when external control of the pH is not being used.

Select the correct option

- phytase
- phosphate
- peptides
- proteins

Click to Save Answer & Move to Next Question

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Time Left sec(s)

BT603 - Fermentation Technology (Quiz No. 2) **Quiz Start Time: 08:12 AM, 07 June 2023**

Question # 9 of 10 (Start time: 08:19:47 AM, 07 June 2023) **Total Marks:** 1

The main product of a fermentation process will often determine the choice of _____ source, particularly if the product results from the direct dissimilation of it.

Select the correct option

- carbon
- nitrogen
- oxygen
- hydrogen

Click to Save Answer & Move to Next Question

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Time Left 80 sec(s)

BT603 - Fermentation Technology (Quiz No. 2)

Quiz Start Time: 08:12 AM, 07 June 2023

Question # 3 of 10 (Start time: 08:15:16 AM, 07 June 2023)

Total Marks: 1

_____ is the major component of all fermentation media, and is needed in many of the ancillary services.

Select the correct option

- water
- carbon
- nitrogen
- hydrogen

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BT603 - Fermentation Technology (Quiz No. 2)

Quiz Start Time: 04:40 PM, 06 June 2023

Question # 6 of 10 (Start time: 04:47:37 PM, 06 June 2023)

Total Marks:

Some proteinaceous nitrogen compounds serving as sources of amino acids include corn-steep liquor, soya meal, peanut meal, _____

Select the correct option

- hydrogen
- oxygen
- carbon
- cotton-seed meal



Click to Save Answer & Move to Next Question

Original text

Contribute a better translation

Question # 4 of 10 (Start time: 08:16:07 AM, 07 June 2023) **Total Marks:** 1

When the ammonium ion has been exhausted during fermentation, there is an alkaline drift as the _____ is used as an alternative nitrogen source.

Select the correct option

- ethanol
- HCL
- nitrate
- citric acid

Click to Save Answer & Move to Next Question

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BT603 - Fermentation Technology (Quiz No. 2)

Time Left 88 sec(s)

Quiz Start Time: 04:40 PM, 06 June 2023

Question # 8 of 10 (Start time: 04:50:35 PM, 06 June 2023)

Total Marks:

Control mechanisms exist by which nitrate reductase, an enzyme involved in the conversion of nitrate to ammonium ion, is repressed in the presence of _____.

Select the correct option

- soya bean
- ammonia
- environmental nitrogen
- Lactobacillus

Click to Save Answer & Move to Next Question

Original text

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Time Left sec(s)

BT603 - Fermentation Technology (Quiz No. 2)

Quiz Start Time: 08:12 AM, 07 June 2023

Question # 5 of 10 (Start time: 08:16:48 AM, 07 June 2023)

Total Marks:

Organic nitrogen may be supplied as amino acid, _____ or urea.

Select the correct option

- carbon
- protein
- hydrogen
- oxygen

Click to Save Answer & Move to Next Question

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Question # 10 of 10 (Start time: 04:36:09 PM, 06 June 2023)

_____ were first used as carriers for antifoams in antibiotic processes.

Select the correct option

- meat
- Oils
- carbohydrates
- proteins

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Question # 8 of 10 (Start time: 04:33:20 PM, 08 June 2023)

There are some advantages and disadvantages of hydrocarbons and their derivatives as fermentation substrates, particularly with reference to ____

Cost

Select the correct option

- cost
- media
- system
- air

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Question # 8 of 10 (Start time: 04:33:20 PM, 06 June 2023)

There are some advantages and disadvantages of hydrocarbons and their derivatives as fermentation substrates, particularly with reference to _____

Select the correct option

- ○ cost
- ○ media
- ○ system
- ○ air

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Question # 9 of 10 (Start time: 04:34:45 PM, 06 June 2023)

Control mechanisms exist by which nitrate reductase, an enzyme involved in the conversion of nitrate to ammonium ion, is repressed in the presence of _____

Select the correct option

- environmental nitrogen
- ammonia
- Lactobacillus
- soya bean

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Question # 7 of 10 (Start time: 04:32:32 PM, 06 June 2023)

In griseofulvin production, adequate available chloride is provided by the inclusion of at least 0.1% _____ as well as the chloride provided by the comple

Select the correct option

- C
- KCl
- Complex media
- N

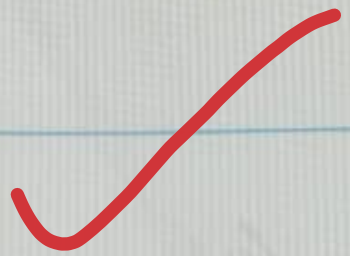
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Question # 6 of 10 (Start time: 04:31:45 PM, 06 June 2023)

_____ is being used as a buffer to minimize pH changes when external control of the pH is not being used.

Select the correct option

- phytase
- proteins
- peptides
- phosphate



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Question # 5 of 10 (Start time: 04:30:58 PM, 06 June 2023)

The use of complex _____ sources for antibiotic production has been common practice.

Select the correct option

- ○ salt
- ○ carbohydrates
- ○ sugar
- ○ nitrogen ✓

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Question # 4 of 10 (Start time: 04:30:06 PM, 06 June 2023)

When the ammonium ion has been exhausted during fermentation, there is an alkaline drift as the _____ is used as an alternative nitroge

Select the correct option

- ethanol
- nitrate
- HCL
- citric acid

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Question # 3 of 10 (Start time: 04:29:04 PM, 06 June 2023)

It is common practice to use _____ as the carbon source in microbial fermentation processes.

Select the correct option

- nitrogen
- carbohydrates
- oxygen
- hydrogen

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Question # 2 of 10 (Start time: 04:28:01 PM, 06 June 2023)

_____ is Best nitrogen sources for some secondary metabolites

Select the correct option

- ○ medicine
- ○ alcohol
- ○ drug
- ○ penicilline ✓

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Time Left sec(s)

BT603 - Fermentation Technology (Quiz No. 2) **Quiz Start Time: 08:12 AM, 07 June 2023**

Question # 2 of 10 (Start time: 08:13:54 AM, 07 June 2023) **Total Marks:**

Ammonium salts such as _____will usually produce acid conditions as the ammonium ion is utilized and the free acid will be liberated.

Select the correct option

- ammonium sulphate
- vitamins
- lipids
- proteins

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Question # 1 of 10 (Start time: 04:26:47 PM, 06 June 2023)

Organic nitrogen may be supplied as amino acid, _____ or urea.

Select the correct option

- protein
- hydrogen
- oxygen
- carbon

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Question # 1 of 10 (Start time: 08:12:37 AM, 07 June 2023) **Total Marks:** 1

Yeast extracts are produced from baker's yeast through autolysis (at 50-55°C) or through plasmolysis (high concentration of _____

not Found

Select the correct option

- NaCl
- alkali
- HCL
- copper sulphate

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Media for Industrial Fermentations

Most industrially used micro-organisms can utilize inorganic or organic sources of nitrogen. Inorganic nitrogen may be supplied as ammonia gas, ammonium salts or nitrates. Ammonia has been used for pH control and as the major nitrogen source in a defined medium for the commercial production of human serum albumin by *Saccharomyces cerevisiae*. Ammonium salts such as ammonium sulphate will usually produce acid conditions as the ammonium ion is utilized and the free acid will be liberated.

On the other hand nitrates will normally cause an alkaline drift as they are metabolized. Ammonium nitrate will first cause an acid drift as the ammonium ion is utilized, and nitrate assimilation is repressed. When the ammonium ion has been exhausted, there is an alkaline drift as the nitrate is used as an alternative nitrogen source.

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drift as the ni 1 of 1

Media for Industrial Fermentations

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❑Oils and Fats

Oils were first used as carriers for antifoams in antibiotic processes. Vegetable oils (olive, maize, cotton seed, linseed, soya bean, etc.) may also be used as carbon substrates, particularly for their content of the fatty acids, oleic, linoleic and linolenic acid, because costs are competitive with those of carbohydrates.

Bader *et al.*, (1984) discussed factors favouring the use of oils instead of carbohydrates. A typical oil contains approximately 2.4 times the energy of glucose on a per weight basis. Oils also have a volume advantage as it would take 1.24 dm^3 of soya bean oil to add 10 kcal of energy to a fermenter, whereas it would take 5 dm^3 of glucose or sucrose assuming that they are being added as 50% w/w solutions.

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Media for Industrial Fermentations

Water is the major component of all fermentation media, and is needed in many of the ancillary services such as

- ✓ heating
- ✓ cooling
- ✓ cleaning
- ✓ rinsing.

When assessing the suitability of a water supply it is important to consider pH, dissolved salts and effluent contamination.

The mineral content of the water is very

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Media for Industrial Fermentations

One exception to this pattern is the metabolism of *Gibberella fujikuroi* (Borrow *et al*., 1961, 1964). In the presence of nitrate the assimilation of ammonia is inhibited at pH 2.8-3.0. Nitrate assimilation continues until the pH has increased enough to allow the ammonia assimilation mechanism to restart.

Organic nitrogen may be supplied as amino acid, protein or urea. In many instances growth will be faster with a supply of organic nitrogen, and a few microorganisms have an absolute requirement for amino acids. Amino acids are more commonly added as complex organic nitrogen sources which are non-homogeneous, cheaper and readily available. In lysine production, methionine and threonine are obtained from soybean hydrolysate since it would be too expensive to use the pure amino acids.

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Media for Industrial Fermentations

The concentration of phosphate in a medium, particularly laboratory media in shake flasks, is often much higher than that of other mineral components. Part of this phosphate is being used as a buffer to minimize pH changes when external control of the pH is not being used.

In specific processes the concentration of certain minerals may be very critical. Some secondary metabolic processes have a lower tolerance range to inorganic phosphate than vegetative growth. This phosphate should be sufficiently low as to be assimilated by the end of trophophase.

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Media for Industrial Fermentations

❑Factors influencing the choice of N_2 source

The use of complex nitrogen sources for antibiotic production has been common practice. They thought to help create physiological conditions in trophophase which favour antibiotic production in idiophase.

For example, in the production of polyene antibiotics, soybean meal has been considered a good nitrogen source because of the balance of nutrients, the low phosphorus content and slow hydrolysis.

It has been suggested that this gradual breakdown prevents the accumulation of ammonium ions and repressive amino acids. These are probably some of the reasons for the selection of ideal nitrogen sources for some secondary metabolites

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ideally 1 of 1

Media for Industrial Fermentations

❑Oils and Fats

Ideally, in any fermentation process, the maximum working capacity of a vessel should be used. Oil based fed-batch fermentations permit this procedure to operate more successfully than those using carbohydrate feeds where a larger spare capacity must be catered for to allow for responses to a sudden reduction in the residual nutrient level Oils also have antifoam properties which may make downstream processing simpler, but normally they are not used solely for this purpose.

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Other proteinaceous nitrogen compounds serving as sources of amino acids include corn-steep liquor, soya meal, peanut meal, cotton-seed meal.

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Media for Industrial Fermentations

❑Oils and Fats

Oils were first used as carriers for antifoams in antibiotic processes. Vegetable oils (olive, maize, cotton seed, linseed, soya bean, etc.) may also be used as carbon substrates, particularly for their content of the fatty acids, oleic, linoleic and linolenic acid, because costs are competitive with those of carbohydrates.

Bader *et al.*, (1984) discussed factors favouring the use of oils instead of carbohydrates. A typical oil contains approximately 2.4 times the energy of glucose on a per weight basis. Oils also have a volume advantage as it would take 1.24 dm^3 of soya bean oil to add 10 kcal of energy to a fermenter, whereas it would take 5 dm^3 of glucose or sucrose assuming that they are being added as 50% w/w solutions.

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